

Affine Jump Intensity for the Two-Factor Stochastic Volatility Model

A complete mathematical development: from point-process foundations
to GMM moments, filtering, and variance-risk-premium analysis

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Abstract

The Step-A calibration of the Todorov (2010) two-factor model was rejected by three held-out diagnostics: the realized jump-variance series JV_t is strongly serially correlated (the model demands independence), the cross-autocovariance between TPV_t and JV_t is two-sided (the model demands strict one-sidedness), and the jump feed-through persists for about a week (the calibrated decay dies in hours). All three failures share one cause: the model's jumps arrive at *constant* intensity. This document develops, from first principles, the minimal repair: an *affine stochastic intensity* $\lambda(t) = \lambda_0 + \lambda_1 V^j(t^-)$, which makes the jump process self-exciting while preserving closed-form moments. We build the point-process machinery from scratch, prove why the affine class is the unique tractable choice, establish the equivalence with an exponential-kernel marked Hawkes process, derive every population moment the GMM estimator needs using Dynkin's formula over fixed time intervals, analyze identification, and trace the consequences through every layer of the project: estimation, the latent-state filter, variance-risk-premium construction, regime classification, and the simulation harness that must verify the algebra before any estimation is run.

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Chapter 1

Orientation: what broke, and what we are building

1.1 The rejected specification

The Step-A model drives price jumps and jump-variance jumps with a single Poisson random measure $\mu(dt, dx)$ whose compensator is $dt G(dx)$: *time-homogeneous*. Consequences that the data rejected:

1. **JV autocovariance.** Under time-homogeneity, jump energies collected over disjoint days are independent, so $\text{Cov}(\text{JV}_t, \text{JV}_{t-k}) = 0$ for every lag $k \geq 1$ and every parameter value. Empirically the lag-1 autocovariance is ≈ 12.4 against a ± 3.5 noise band, decaying slowly through lag ~ 10 . *Jumps cluster*.
2. **Cross-autocovariance asymmetry.** The model implies $\text{Cov}(\text{TPV}_t, \text{JV}_{t+k}) = 0$ for $k \geq 1$: past diffusive variance cannot predict future jump energy, because future jump arrivals are independent of everything. Empirically that branch is clearly positive out to $k \approx 7$.
3. **Feed-through persistence.** The calibrated jump-factor decay $|\hat{\rho}_j| = 6.14$ (half-life 0.11 d) kills the model-implied feed-through of jump energy into future TPV within two days; empirically it persists about a week.

1.2 The repair in one paragraph

Let the arrival rate of jumps be an *affine function of the jump-variance state itself*:

$$\lambda(t) = \lambda_0 + \lambda_1 V^j(t^-), \quad \lambda_0 > 0, \lambda_1 \geq 0.$$

Every jump raises V^j , which raises the arrival rate of *further* jumps: a self-exciting cascade. One new parameter (λ_1) simultaneously produces (i) exponentially decaying JV autocovariance, (ii) a two-sided cross-autocovariance whose forward branch exceeds its backward branch by exactly the co-jump coupling, and (iii) an *effective* persistence rate κ_j that is *slower* than the per-jump decay rate $b = -\rho_j$ — resolving the paradox of fast individual decay coexisting with slow observed persistence. Because the intensity is affine, all unconditional and conditional moments remain in closed form, so the entire GMM/filtering/VRP pipeline survives with formulas swapped, not rebuilt.

1.3 How this document is organized

Chapter 2 builds point processes, random measures, compensators and the martingale calculus we need — from zero. Chapter 3 explains what “affine” means, proves why it is the tractability frontier, and identifies our specification with a marked Hawkes process. Chapter 4 states the extended model and proves the central effective-rate theorem. Chapter 5 is the moment engine: every population quantity the estimator needs, derived with Dynkin’s formula over fixed intervals, each ending in a boxed formula. Chapter 6 treats identification and the GMM design. Chapter 7 rebuilds the latent-state projection. Chapter 8 connects the calibrated physical dynamics to the variance risk premium and regime classification. Chapter 9 specifies simulation, verification, and the concrete changes to `test.py`. Appendices collect derivatives for the analytic Jacobian, useful integrals, notation, and an implementation checklist.

Mathematical house rules. Per the project’s estimation protocol, all derivations proceed through *Dynkin’s formula evaluated over a fixed global time interval*. We never shrink a time step to zero as a derivation device and we never invoke the adjoint (forward/Fokker–Planck) operator. Stationary moments follow from the fixed-interval identity $\mathbb{E} \int_0^T \mathcal{A}f(X_s) \, ds = \mathbb{E}f(X_T) - \mathbb{E}f(X_0) = 0$; conditional moments follow from the same identity read as a Volterra integral equation in the interval endpoint.

Chapter 2

Point processes, random measures, and compensators

2.1 Counting processes and the Poisson process

Definition 2.1 (Counting process). A *counting process* $N = (N_t)_{t \geq 0}$ is a right-continuous, nondecreasing, integer-valued process with $N_0 = 0$ and unit jumps. N_t counts events in $(0, t]$.

Definition 2.2 (Poisson process). N is a *Poisson process with rate $\lambda > 0$* if it has independent increments and $N_t - N_s \sim \text{Poisson}(\lambda(t - s))$ for $s < t$.

Two properties matter for us. First, $N_t - \lambda t$ is a martingale: the “compensated” process has no predictable trend. Second, independent increments are exactly what the data rejected — event counts (and energies) in disjoint windows are uncorrelated. Everything in this chapter generalizes these two facts to state-dependent rates.

2.2 Marked point processes and random measures

Our jumps carry a *mark* x (the jump size information): the i -th event is a pair (τ_i, x_i) of a time and a mark in $\mathbb{R}_0 = \mathbb{R} \setminus \{0\}$ (or a more general mark space; nothing below depends on it). The bookkeeping device is the *random counting measure*

$$\mu(A \times B) = \#\{i : \tau_i \in A, x_i \in B\}, \quad A \subseteq [0, \infty), B \subseteq \mathbb{R}_0,$$

so that for any function ψ of the mark,

$$\int_0^t \int_{\mathbb{R}_0} \psi(x) \mu(ds, dx) = \sum_{i: \tau_i \leq t} \psi(x_i)$$

is the running sum of ψ over the jumps so far. Realized jump variance is exactly such a sum with $\psi = h^2$; the jump-variance factor V^j is driven by such a sum with $\psi = k$.

2.3 Stochastic intensity and the compensator

Definition 2.3 (Stochastic intensity). Let $(\mathcal{F}_t)$ be the information filtration. A nonnegative, *predictable* process $\lambda(t)$ is the $(\mathcal{F}_t)$ -*intensity* of the marked point process μ with i.i.d. marks

$x_i \sim G$ if

$$\nu(dt, dx) = \lambda(t) dt G(dx)$$

compensates μ : for every bounded predictable integrand ψ ,

$$M_t = \int_0^t \int \psi(x) [\mu(ds, dx) - \nu(ds, dx)] \quad \text{is an } (\mathcal{F}_t)\text{-martingale.} \quad (2.1)$$

We write $\tilde{\mu} = \mu - \nu$ for the compensated measure. We normalize G to be a *probability* measure, so $\lambda(t)$ is the arrival *rate* of jumps and marks are i.i.d. draws from G .

Intuition: over the fixed window $[t, t + \Delta]$ the expected number of jumps, given everything known at t and along the path, is $\mathbb{E} \int_t^{t+\Delta} \lambda(s) ds$; the compensator is the “predictable budget” of jumps, and subtracting it leaves pure surprise. Predictability of λ (using $V^j(t^-)$, the left limit) is not pedantry: the rate at which the next jump arrives must be determined by the state *just before* it arrives, otherwise a jump could raise its own arrival probability retroactively and (2.1) would fail.

Remark 2.4 (Normalization choice). The Step-A code used the Lévy-measure convention, absorbing the rate into G : its parameters $\mathcal{I}_{h^2} = \int h^2 dG^{\text{old}}$ etc. are, in the present convention, the products $\mathcal{I}_{h^2} = \lambda m_{h^2}$, $\mathcal{I}_{h^4} = \lambda m_{h^4}$, $\mathcal{I}_{kh^2} = \lambda m_{kh^2}$, where $m_\phi := \int \phi(x) G(dx)$. Keeping rate and mark distribution separate is what lets the rate become stochastic. The dictionary between old and new parameters is given in Table 9.1.

2.4 The moment toolkit for compensated integrals

The three formulas below are the entire “Itô calculus” this project needs for jumps. Fix a day $[t-1, t]$ and write $M[\psi] = \int_{t-1}^t \int \psi(x) \tilde{\mu}(ds, dx)$.

Lemma 2.5 (First moments). $\mathbb{E} \int_{t-1}^t \int \psi(x) \mu(ds, dx) = m_\psi \mathbb{E} \int_{t-1}^t \lambda(s) ds$, and $\mathbb{E} M[\psi] = 0$.

Lemma 2.6 (Itô isometry for marked point processes). *For predictable ψ, ϕ ,*

$$\mathbb{E}(M[\psi] M[\phi]) = m_{\psi\phi} \mathbb{E} \int_{t-1}^t \lambda(s) ds.$$

In particular $\text{Var}(M[\psi]) = m_{\psi^2} \mathbb{E} \int \lambda$.

Lemma 2.7 (Orthogonality of increments). *If Z is $\mathcal{F}_s$ -measurable and $M[\psi]$ is a compensated integral over $(s, t]$, then $\mathbb{E}[Z M[\psi]] = 0$. In particular, compensated jump integrals over disjoint intervals are uncorrelated, and a compensated integral is uncorrelated with any functional of the earlier path.*

Why these hold. Lemma 2.5 is the definition of compensation applied to ψ and the tower property. Lemma 2.6: jumps occur at distinct times, so the product of two jump sums over the same window is a sum over *common* jumps of $\psi(x_i)\phi(x_i)$ plus cross terms of distinct jumps whose compensated expectations vanish; the common-jump sum has expectation $m_{\psi\phi} \mathbb{E} \int \lambda$ by Lemma 2.5. Lemma 2.7 is the martingale property plus the tower property: $\mathbb{E}[Z M] = \mathbb{E}[Z \mathbb{E}(M | \mathcal{F}_s)] = 0$. $\square$

These lemmas are deliberately stated with a *stochastic* λ : they are exactly where the new model’s extra covariance terms will come from, because $\mathbb{E} \int \lambda$ ceases to be a constant and starts to covary with the volatility path.

Chapter 3

The affine principle, and why it is the tractability frontier

3.1 Definition and the closure property

Definition 3.1 (Affine intensity). The intensity is *affine* in the state if $\lambda(t) = \lambda_0 + \lambda_1 V^j(t^-)$ with $\lambda_0 > 0$, $\lambda_1 \geq 0$ (and the state dynamics keep $V^j \geq 0$, which holds here because V^j decays exponentially between nonnegative jumps).

The deep reason to insist on affinity is an algebraic closure property of the *extended generator*. For our jump factor

$$dV^j(t) = \rho_j V^j(t) dt + \int_{\mathbb{R}_0} k(x) \mu(dt, dx), \quad b := -\rho_j > 0, \quad k \geq 0,$$

the generator acts on a smooth f as

$$\mathcal{A}f(v) = -bv f'(v) + (\lambda_0 + \lambda_1 v) \int_{\mathbb{R}_0} [f(v + k(x)) - f(v)] G(dx). \quad (3.1)$$

The first term is the deterministic decay; the second says: at rate $\lambda(v)$, replace v by $v + k(x)$ with $x \sim G$.

Proposition 3.2 (Polynomial closure). *If $\lambda(v)$ is affine, then $\mathcal{A}$ maps polynomials of degree n to polynomials of degree n . If $\lambda(v)$ is a polynomial of degree $d \geq 2$, then $\mathcal{A}$ maps degree n to degree $n + d - 1 > n$.*

Proof. Apply (3.1) to $f(v) = v^n$. The drift term gives $-nbv^n$: degree n . The jump term gives $\lambda(v) \sum_{r=0}^{n-1} \binom{n}{r} v^r m_{k^{n-r}}$, a polynomial of degree $(n-1) + \deg \lambda$. With $\deg \lambda = 1$ this is degree n ; with $\deg \lambda = d$ it is degree $n + d - 1$. $\square$

Project impact 3.3. Proposition 3.2 is why the whole project works. Moment derivations reduce (Chapter 5) to computing $\mathbb{E}[\mathcal{A}f]$ for polynomial f . Under affinity, the n -th moment equation involves only moments of order $\leq n$: a finite, triangular linear system — closed form, hence an analytic GMM Jacobian, hence stable optimization. Under a quadratic intensity, the second-moment equation would involve the third moment, the third the fourth, ad infinitum: no closed form, forcing simulation-based estimation (orders of magnitude slower, noisy gradients, no analytic Jacobian — the exact failure modes the project's coding standards prohibit). Affinity

is not a stylistic choice; it is the boundary between “rederive formulas in an afternoon” and “rebuild the estimator.”

3.2 Equivalence with a marked Hawkes process

Proposition 3.4 (Hawkes representation). *Suppose V^j starts from its stationary distribution in the infinite past. Then*

$$V^j(t) = \sum_{\tau_i \leq t} k(x_i) e^{-b(t-\tau_i)}, \quad \lambda(t) = \lambda_0 + \lambda_1 \sum_{\tau_i < t} k(x_i) e^{-b(t-\tau_i)}.$$

That is: the pair (jump times, intensity) is precisely a marked Hawkes process with exponential kernel $\lambda_1 k(x) e^{-bu}$ and baseline λ_0 .

Proof. Between jumps V^j satisfies $dV^j = -bV^j dt$, so each jump’s contribution $k(x_i)$ decays as $e^{-b(t-\tau_i)}$; summing contributions gives the display (the transient from the infinite past vanishes). Substituting into $\lambda = \lambda_0 + \lambda_1 V^j(t^-)$ gives the Hawkes form. $\square$

The Hawkes lens supplies the two most useful intuitions in this document.

Cluster (branching) representation. A Hawkes process is statistically identical to the following cascade: *immigrant* events arrive as a plain Poisson process with rate λ_0 ; each event (of any generation) with mark x independently spawns *children* as an inhomogeneous Poisson process with rate $\lambda_1 k(x) e^{-bu}$, u its age. The expected number of children per event is

$$\eta = \int_0^\infty \lambda_1 m_k e^{-bu} du = \frac{\lambda_1 m_k}{b} \quad (\text{the branching ratio}). \quad (3.2)$$

A cluster started by one immigrant has expected total size $1 + \eta + \eta^2 + \dots = (1 - \eta)^{-1}$: finite iff $\eta < 1$.

Assumption 3.5 (Subcriticality). $\eta = \lambda_1 m_k / b < 1$. This is necessary and sufficient for a stationary version of (V^j, λ) with finite moments of all orders used below.

Two decay rates. Each individual excitation decays at rate b (fast, hours). But a burst is a *cluster*: children arrive while the parent’s excitation wanes, grandchildren while the children’s wanes. The aggregate relaxation of the cascade is slower than any individual decay. The next chapter makes this exact: the observable persistence rate is $\kappa_j = b(1 - \eta)$, and $\eta \rightarrow 1$ drags $\kappa_j \rightarrow 0$ regardless of how fast b is. This is the mathematical resolution of the Step-A paradox (fitted $|\rho_j| = 6.14$ versus week-long empirical feed-through): the old model owns one decay rate and was forced to compromise; the new model owns two.

Chapter 4

The extended model

4.1 Full specification

Assumption 4.1 (Model $\mathcal{M}'$). Under the physical measure $\mathbb{P}$, on a filtered probability space carrying a Brownian motion B_1 and a marked point process μ with i.i.d. marks $x \sim G$ (a probability measure) and stochastic intensity $\lambda(t)$:

$$dV^c(t) = \kappa_c(\bar{V}^c - V^c(t)) dt + \sigma_{cv}\sqrt{V^c(t)} dB_1(t), \quad (\text{CIR, unchanged}) \quad (4.1)$$

$$dV^j(t) = \rho_j V^j(t) dt + \int_{\mathbb{R}_0} k(x) \mu(dt, dx), \quad b := -\rho_j > 0, \quad (4.2)$$

$$\lambda(t) = \lambda_0 + \lambda_1 V^j(t^-), \quad \lambda_0 > 0, \lambda_1 \geq 0, \quad (4.3)$$

with $B_1 \perp \mu$, $k \geq 0$, and Assumption 3.5. Total diffusive variance is $\sigma^2(t) = V^c(t) + V^j(t)$; the same jumps that feed V^j carry price-jump energy $h^2(x)$. The daily observables over $[t-1, t]$ are

$$\text{TPV}_t \approx \int_{t-1}^t (V^c(s) + V^j(s)) ds, \quad \text{JV}_t \approx \int_{t-1}^t \int h^2(x) \mu(ds, dx).$$

The translation gauge $k(x) = c h^2(x)$, $c > 0$, links the two jump kernels, so mark moments satisfy $m_k = c m_{h^2}$, $m_{k^2} = c^2 m_{h^4}$, $m_{kh^2} = c m_{h^4}$.

Setting $\lambda_1 = 0$ recovers the Step-A model exactly: the extension *nests* the old one, which is essential — it lets the data vote on λ_1 , and it gives a free correctness check on every new formula (each must reduce to the old one at $\lambda_1 = 0$).

4.2 The effective persistence rate

Theorem 4.2 (Stationary mean and effective rate). *Under Assumptions 3.5–4.1, the stationary means are*

$$\bar{\lambda} := \mathbb{E}[\lambda(t)] = \frac{\lambda_0}{1 - \eta}, \quad \bar{v} := \mathbb{E}[V^j(t)] = \frac{\bar{\lambda} m_k}{b} = \frac{\lambda_0 m_k}{b(1 - \eta)},$$

and the conditional mean relaxes exponentially at the effective rate

$$\boxed{\kappa_j = b - \lambda_1 m_k = b(1 - \eta)} \quad \text{i.e.} \quad \mathbb{E}[V^j(t+s) | \mathcal{F}_t] = \bar{v} + (V^j(t) - \bar{v})e^{-\kappa_j s}.$$

Consequently $\text{Cov}(V^j(t), V^j(t+s)) = \text{Var}(V^j) e^{-\kappa_j |s|}$.

Proof. Apply Dynkin's formula on the fixed interval $[t, t + s]$ to $f(v) = v$, for which (3.1) gives $\mathcal{A}f(v) = -bv + (\lambda_0 + \lambda_1 v) m_k$:

$$\mathbb{E}_t[V^j(t + s)] - V^j(t) = \mathbb{E}_t \int_t^{t+s} \left[-b V^j(u) + m_k(\lambda_0 + \lambda_1 V^j(u)) \right] du.$$

Write $g(s) = \mathbb{E}_t[V^j(t + s)]$. The identity above, read for every endpoint s of the fixed interval, is the Volterra equation $g(s) = g(0) + \int_0^s [\lambda_0 m_k - \kappa_j g(u)] du$ with $\kappa_j = b - \lambda_1 m_k$; its unique solution is the boxed exponential with resting point $\lambda_0 m_k / \kappa_j = \bar{v}$. Stationary means: take $f(v) = v$ in the stationarity identity $\mathbb{E}[\mathcal{A}f] = 0$ to get $b\bar{v} = \bar{\lambda} m_k$, and take expectations in (4.3) to get $\bar{\lambda} = \lambda_0 + \lambda_1 \bar{v}$; solving the pair gives the display. The autocovariance follows from the conditional mean by the tower property: $\text{Cov}(V_t^j, V_{t+s}^j) = \mathbb{E}[(V_t^j - \bar{v}) \mathbb{E}_t(V_{t+s}^j - \bar{v})] = \text{Var}(V^j) e^{-\kappa_j s}$. $\square$

Remark 4.3 (Reading the theorem). b is the *microscopic* decay: how fast one excitation wanes. $\kappa_j = b(1 - \eta)$ is the *macroscopic* decay: how fast the whole cascade relaxes, slowed by the factor $(1 - \eta)$ because decaying excitation keeps being replenished by offspring jumps. The data constrains κ_j (through every autocovariance and cross-covariance decay), while b appears separately only through the mean level $\bar{v} = \bar{\lambda} m_k / b$. The Step-A transience mandate (“jump factor half-life under 2 days”) properly applies to b ; the persistence the diagnostics revealed lives in κ_j . Both can now be honored at once.

Theorem 4.4 (Stationary variance). $S := \text{Var}(V^j) = \frac{\bar{\lambda} m_k^2}{2\kappa_j}$.

Proof. Take $f(v) = v^2$: $\mathcal{A}f(v) = -2bv^2 + (\lambda_0 + \lambda_1 v)(2vm_k + m_{k^2})$. Stationarity ($\mathbb{E}[\mathcal{A}f] = 0$) gives

$$2(b - \lambda_1 m_k) \mathbb{E}[V^2] = 2\lambda_0 m_k \bar{v} + m_{k^2} \bar{\lambda}.$$

From the proof of Theorem 4.2, $\lambda_0 m_k = \kappa_j \bar{v}$, so $\mathbb{E}[V^2] = \bar{v}^2 + \bar{\lambda} m_{k^2} / (2\kappa_j)$, and subtracting $\bar{v}^2$ gives the claim. $\square$

Note the same effective rate κ_j divides the variance: stronger self-excitation ($\eta \uparrow$) both slows the decay and fattens the stationary spread — one mechanism, two visible symptoms, which is why the data can identify it.

Chapter 5

The moment engine

This chapter derives every population moment used downstream. Throughout, $S = \text{Var}(V^j)$, $\kappa_j = b(1 - \eta)$, $\bar{\lambda}$, $\bar{v}$ are as in Chapter 4, and $\sigma_c^2 = \sigma_{cv}^2 \bar{V}^c / (2\kappa_c)$ is the CIR stationary variance (unchanged from Step A).

5.1 The damping trio

Daily observables are *time integrals*, which blur point-in-time covariances. All blurring is captured by three scalar functions.

Lemma 5.1 (Integration damping). *Let X be stationary with $\text{Cov}(X_t, X_{t+s}) = \varsigma^2 e^{-a|s|}$, and let $I_t = \int_{t-1}^t X_s ds$. Define*

$$f(a) = \frac{2(a - 1 + e^{-a})}{a^2}, \quad \hat{g}(a) = \frac{1 - e^{-a}}{a}, \quad g(a) = \hat{g}(a)^2 = \frac{(1 - e^{-a})^2}{a^2}.$$

Then for integer $k \geq 1$:

$$\text{Var}(I_t) = \varsigma^2 f(a), \quad \text{Cov}(I_t, I_{t-k}) = \varsigma^2 g(a) e^{-a(k-1)}, \quad \text{Cov}(X_t, I_{t-k}) = \varsigma^2 \hat{g}(a) e^{-ak} \quad (k \geq 0).$$

Proof. All three are Fubini integrals of $e^{-a|u-w|}$ over, respectively, the unit square $\{u, w \in [0, 1]\}$ (split along the diagonal, giving f), the displaced square $\{u \in [k, k+1], w \in [0, 1]\}$ (giving $g e^{-a(k-1)}$), and the segment $\{w \in [k, k+1]\}$ with u fixed at 0 distance k away (giving $\hat{g} e^{-ak}$). The piecewise transition $f \neq g$ at the $k=0 \rightarrow k=1$ boundary is the structural discontinuity the spec calls the “time-integration blur.” $\square$

Numerically, $f, g, \hat{g}$ must be evaluated with `expm1` to avoid catastrophic cancellation at small a (Appendix A collects values, limits, and derivatives).

5.2 The covariance kernel of jumps with the future state

The single computation from which all cross moments flow:

Lemma 5.2 (Feed-through kernel). *For $s \geq u$, with $M[\psi]$ the compensated ψ -jump integral,*

$$\text{Cov}(\psi\text{-jump surprise in } du, V^j(s)) = \bar{\lambda} m_{k\psi} e^{-\kappa_j(s-u)} du,$$

and the covariance is zero for $s < u$. In integrated form, for the running martingale $M_s = \int_0^s \psi \, d\tilde{\mu}$ started at 0 from stationarity, $\text{Cov}(V^j(s), M_s) = \frac{\bar{\lambda} m_{k\psi}}{\kappa_j} (1 - e^{-\kappa_j s})$.

Proof. Apply Dynkin's formula on the fixed interval $[0, s]$ to the pair $(V, M) = (V^j, M)$ with $f(v, m) = vm$. Between jumps, $dV = -bV \, dt$ and $dM = -\lambda(t)m_\psi \, dt$ (the compensator drift); a jump with mark x moves $(v, m) \mapsto (v + k(x), m + \psi(x))$. Hence

$$\mathcal{A}f(v, m) = -bvm - \lambda(v)m_\psi v + \lambda(v)[vm_\psi + mm_k + m_{k\psi}] = -bvm + \lambda(v)(mm_k + m_{k\psi}).$$

Take expectations with $\phi(s) := \mathbb{E}[V(s)M(s)]$, use $\mathbb{E}[M] = 0$ (so $\mathbb{E}[\lambda M] = \lambda_1 \mathbb{E}[VM]$) and $\mathbb{E}[\lambda] = \bar{\lambda}$:

$$\phi(s) = \int_0^s [-\kappa_j \phi(u) + \bar{\lambda} m_{k\psi}] \, du \implies \phi(s) = \frac{\bar{\lambda} m_{k\psi}}{\kappa_j} (1 - e^{-\kappa_j s}),$$

again a Volterra identity on a fixed interval. Differentiating the closed form in s and matching against $\phi(s) = \int_0^s (\text{kernel})(s, u) \, du$ identifies the kernel as $\bar{\lambda} m_{k\psi} e^{-\kappa_j(s-u)}$. For $s < u$ Lemma 2.7 gives zero. $\square$

Note carefully: the kernel decays at κ_j , *not* b . A jump's direct excitation decays at b , but its *descendants* keep the expectation elevated; the conditional-mean rate κ_j governs everything observable.

5.3 Decomposing JV

The master decomposition, used everywhere below:

$$\text{JV}_t = \underbrace{M_t}_{\text{jump surprise}} + m_{h^2} \Lambda_t, \quad M_t := \int_{t-1}^t \int h^2 \, d\tilde{\mu}, \quad \Lambda_t := \int_{t-1}^t \lambda(s) \, ds = \lambda_0 + \lambda_1 \int_{t-1}^t V^j(s) \, ds. \quad (5.1)$$

M_t is white (Lemma 2.7); all serial dependence of JV rides on Λ_t , i.e. on the integrated V^j path — which is also what TPV observes. This one line already explains, qualitatively, every diagnostic: JV autocorrelates because Λ does; TPV predicts future JV because both load on V^j ; and jump surprises feed future TPV through the kernel of Lemma 5.2.

5.4 The complete moment set

Write $\mathcal{E}_k := e^{-\kappa_j(k-1)}$ and $\mathcal{E}_k^c := e^{-\kappa_c(k-1)}$ for $k \geq 1$.

Theorem 5.3 (Population moments of the observables). *Under Assumption 4.1, for lags $k \geq 1$:*

$$\textbf{Means:} \quad \mathbb{E}[\text{TPV}_t] = \bar{V}^c + \bar{v}, \quad \mathbb{E}[\text{JV}_t] = \bar{\lambda} m_{h^2}. \quad (5.2)$$

$$\textbf{TPV block:} \quad \text{Var}(\text{TPV}_t) = \sigma_c^2 f(\kappa_c) + S f(\kappa_j), \quad (5.3)$$

$$\text{Cov}(\text{TPV}_t, \text{TPV}_{t-k}) = \sigma_c^2 g(\kappa_c) \mathcal{E}_k^c + S g(\kappa_j) \mathcal{E}_k. \quad (5.4)$$

$$\textbf{JV block:} \quad \text{Var}(\text{JV}_t) = \bar{\lambda} m_{h^4} + \left[\lambda_1^2 m_{h^2}^2 S + \lambda_1 m_{h^2} \bar{\lambda} m_{kh^2} \right] f(\kappa_j), \quad (5.5)$$

$$\text{Cov}(\text{JV}_t, \text{JV}_{t-k}) = \underbrace{\lambda_1 m_{h^2} [\lambda_1 m_{h^2} S + \bar{\lambda} m_{kh^2}]}_{=:C_J} g(\kappa_j) \mathcal{E}_k. \quad (5.6)$$

$$\textbf{Cross block:} \quad \text{Cov}(\text{TPV}_t, \text{JV}_t) = \bar{\lambda} m_{kh^2} \frac{1}{2} f(\kappa_j) + \lambda_1 m_{h^2} S f(\kappa_j), \quad (5.7)$$

$$\text{Cov}(\text{TPV}_t, \text{JV}_{t-k}) = \underbrace{[\bar{\lambda} m_{kh^2} + \lambda_1 m_{h^2} S]}_{=:C_F} g(\kappa_j) \mathcal{E}_k \quad (\text{jumps lead}). \quad (5.8)$$

$$\text{Cov}(\text{TPV}_t, \text{JV}_{t+k}) = \underbrace{\lambda_1 m_{h^2} S}_{=:C_B} g(\kappa_j) \mathcal{E}_k \quad (\text{variance leads}). \quad (5.9)$$

Proof. Means: Lemma 2.5 with $\psi = h^2$ and Theorem 4.2. *TPV block:* $V^c \perp V^j$, each with an exponential autocovariance (κ_c from the CIR, κ_j from Theorem 4.2); apply Lemma 5.1 factorwise. *JV block:* expand (5.1). $\text{Var}(M_t) = \bar{\lambda} m_{h^4}$ (Lemma 2.6); $\text{Var}(m_{h^2} \Lambda_t) = m_{h^2}^2 \lambda_1^2 S f(\kappa_j)$ (Lemma 5.1 applied to V^j); $2m_{h^2} \text{Cov}(M_t, \Lambda_t) = 2m_{h^2} \lambda_1 \text{Cov}(M_t, \int V^j) = 2m_{h^2} \lambda_1 \cdot \bar{\lambda} m_{kh^2} \cdot \frac{1}{2} f(\kappa_j)$, where the last step integrates the kernel of Lemma 5.2 ($\psi = h^2$, $m_{kh^2} = m_{kh^2}$) over the triangle $\{u \leq s\}$ of the unit square — exactly the $\frac{1}{2}f$ geometry of Lemma 5.1. Summing gives (5.5). For lags: M 's of disjoint days are mutually orthogonal and orthogonal to the earlier path (Lemma 2.7), so only Λ - Λ and past- M -future- Λ terms survive: $m_{h^2}^2 \lambda_1^2 S g(\kappa_j) \mathcal{E}_k$ and $m_{h^2} \lambda_1 \cdot \bar{\lambda} m_{kh^2} g(\kappa_j) \mathcal{E}_k$ (kernel integrated over displaced squares); factoring yields (5.6). *Cross block:* $\text{Cov}(\int V^j\text{-day } t, \text{JV}_{t-k})$ has an M -part (kernel over the displaced square: $\bar{\lambda} m_{kh^2} g(\kappa_j) \mathcal{E}_k$) and a Λ -part ($\lambda_1 m_{h^2} S g(\kappa_j) \mathcal{E}_k$), giving (5.8); at $k = 0$ the M -part sees only the triangle ($\frac{1}{2}f$) while the Λ -part sees the full square (f), giving (5.7); in the backward direction the M -part dies (future surprises are orthogonal to the past, Lemma 2.7) and only the Λ -part survives, giving (5.9). V^c contributes nowhere in the JV/cross blocks by independence. $\square$

Corollary 5.4 (Nesting). *At $\lambda_1 = 0$: $\kappa_j = b$, $\bar{\lambda} = \lambda_0$, $C_J = C_B = 0$, $C_F = \lambda_0 m_{kh^2}$, and every formula collapses to the Step-A model with $\mathcal{I}_{h^2} = \lambda_0 m_{h^2}$, $\mathcal{I}_{h^4} = \lambda_0 m_{h^4}$, $\mathcal{I}_{kh^2} = \lambda_0 m_{kh^2}$.*

Corollary 5.5 (Structure of the extension). *The three rejected diagnostics are repaired with signs and shapes forced by the theory:*

1. JV autocovariance is a single exponential in κ_j with positive weight C_J — matching the observed slow decay;
2. the cross-autocovariance is two-sided, and its asymmetry is exactly the co-jump coupling: $C_F - C_B = \bar{\lambda} m_{kh^2} \geq 0$ — matching the observed “both sides positive, forward larger”;
3. all serial decay happens at the effective rate κ_j , which can be slow (persistent cascades) even while b is fast (transient individual shocks) — matching the week-long feed-through.

Moreover the three coefficients obey the *internal consistency identity*

$$C_J \cdot S = C_B \cdot C_F$$

(immediate from $C_J = \lambda_1 m_{h^2} C_F$ and $C_B = \lambda_1 m_{h^2} S$): an over-identifying restriction imposed on empirical quantities that the data can reject. It is a free, sharp specification test and should be reported with the calibration.

Chapter 6

Identification and GMM design

6.1 What is actually identified: composite parameters

Inspect Theorem 5.3: the raw quantities $(\lambda_0, \lambda_1, m_{h^2}, m_{h^4}, c)$ never appear alone. Every moment is a function of the *composites*

$$A := \bar{\lambda} m_{h^2}, \quad B := \bar{\lambda} m_{h^4}, \quad \psi := \lambda_1 m_{h^2}, \quad c, \quad \kappa_j, \quad \kappa_c, \quad \sigma_c^2, \quad \bar{V}^c, \quad S,$$

using the gauge to write $m_{kh^2} = c m_{h^4}$, hence $\bar{\lambda} m_{kh^2} = cB$, and (from Theorem 4.2)

$$\bar{v} = \frac{\bar{\lambda} m_k}{b} = \frac{cA}{b}, \quad b = \kappa_j + \psi c, \quad \eta = \frac{\psi c}{b}.$$

In composite form the moment set reads:

$$\begin{aligned} \mathbb{E}[\text{JV}] &= A, & \text{Var}(\text{JV}) &= B + (\psi^2 S + \psi cB) f(\kappa_j), \\ \mathbb{E}[\text{TPV}] &= \bar{V}^c + \frac{cA}{\kappa_j + \psi c}, & \gamma_k^{\text{JV}} &= \psi (cB + \psi S) g(\kappa_j) \mathcal{E}_k, \\ \text{Var}(\text{TPV}) &= \sigma_c^2 f(\kappa_c) + S f(\kappa_j), & \gamma_k^{\text{TPV}} &= \sigma_c^2 g(\kappa_c) \mathcal{E}_k^c + S g(\kappa_j) \mathcal{E}_k, \\ \text{Cov}(\text{TPV}_t, \text{JV}_t) &= cB \frac{f(\kappa_j)}{2} + \psi S f(\kappa_j), & \text{cross}_k^\pm &= (cB + \psi S) \text{ or } (\psi S) \times g(\kappa_j) \mathcal{E}_k. \end{aligned}$$

Proposition 6.1 (Scale degeneracy and normalization). *$(\lambda_0, \lambda_1, m_{h^2}, m_{h^4})$ enter only through (A, B, ψ) : the split of jump activity between rate and mark size is not identified from these moments (“twice as many jumps, half the energy each” is observationally equivalent at this moment order). We therefore normalize the mark scale, $m_{h^2} := 1$, after which*

$$\bar{\lambda} = A, \quad m_{h^4} = B/A, \quad \lambda_1 = \psi, \quad \lambda_0 = \bar{\lambda}(1 - \eta)$$

are exactly recovered. This is a labeling convention, not a restriction: every observable quantity and every downstream object (filter, VRP forecast) depends on the composites only.

Remark 6.2 (Gauge: strict vs. relaxed). The gauge $k = c h^2$ applied to *second* mark moments forces $m_{k^2} = c^2 m_{h^4}$, which by Theorem 4.4 ties down $S = c^2 B / (2\kappa_j)$ — removing S from the free set. We recommend estimating with S **free** (“relaxed gauge”: the gauge is used only where the spec requires it, for the level recovery $\bar{v} = cA/b$), for two reasons: it nests the Step-A treatment (where $\text{Var}(V^j)$ was free), and it converts the strict-gauge equation into a second testable over-identifying restriction, $S \stackrel{?}{=} c^2 B / (2\kappa_j)$, to report alongside $C_J S = C_B C_F$. If both restrictions survive, re-estimate under the strict gauge for efficiency.

6.2 The estimable vector, bounds, and moment menu

Parameter vector (relaxed gauge, $m_{h^2} = 1$):

$$\Theta' = [\kappa_c, \sigma_c^2, \bar{V}^c, \kappa_j, A, B, \psi, c, S] \in \mathbb{R}^9.$$

Derived (reported, not estimated): $b = \kappa_j + \psi c$, $\eta = \psi c/b$, $\bar{\lambda} = A$, $\lambda_0 = A(1 - \eta)$, $\lambda_1 = \psi$, $\bar{v} = cA/b$, $\rho_j = -b$.

Bounds. All of $\sigma_c^2, \bar{V}^c, A, B, \psi, c, S$ strictly positive; κ_c in its persistence band as mandated. Two structural constraints deserve emphasis:

- **Stationarity is free.** $\eta < 1 \iff \kappa_j > 0$: by parametrizing with κ_j directly, subcriticality is enforced by a simple positivity bound — no nonlinear constraint needed. This is a real design win of the composite parametrization.
- **The old transience mandate moves to b .** “Jump factor half-life < 2 days” constrains $b = \kappa_j + \psi c \geq \ln 2/2$, a *linear* inequality in (κ_j, ψ, c) -space (impose via `scipy.optimize` linear constraint or check-and-report). κ_j itself must be left *free to be persistent* — that freedom is the entire point of the extension.

Moment menu. The recommended GMM target set (all formulas from Theorem 5.3):

block	moments	count
means	$\mathbb{E}[\text{TPV}], \mathbb{E}[\text{JV}]$	2
variances	$\text{Var}(\text{TPV}), \text{Var}(\text{JV})$	2
contemporaneous cross	$\text{Cov}(\text{TPV}_t, \text{JV}_t)$	1
TPV autocovariance	$k = 1, \dots, 15$	15
JV autocovariance	$k = 1, \dots, 10$	10
forward cross (jumps lead)	$k = 1, \dots, 5$	5
backward cross (variance leads)	$k = 1, \dots, 5$	5
total		40

Nine parameters, forty moments: heavily over-identified, by design — the over-identification *is* the specification test. Note the reversal from Step A: the JV autocovariances, previously excluded because their Jacobian rows were identically zero, are now the *primary* carriers of information about ψ and κ_j ; the backward cross moments are the only block that separates $C_B = \psi S$ from $C_F = cB + \psi S$, i.e. they identify c .

Who identifies whom (the mental map).

parameter	primary identifying feature
κ_j	common decay slope of JV-acov, cross-acov (log-linear in k)
κ_c, σ_c^2	slow residual component of TPV-acov after the κ_j part
S	fast component weight in TPV-acov ($Sg(\kappa_j)$)
A	$\mathbb{E}[\text{JV}]$
B	$\text{Var}(\text{JV})$ level (the f -terms are second order)
ψ	JV-acov weight C_J and backward-cross weight $C_B = \psi S$
c	cross asymmetry $C_F - C_B = cB$
$\bar{V}^c$	$\mathbb{E}[\text{TPV}]$ net of $\bar{v} = cA/b$

6.3 Weighting, conditioning, and the Jacobian

Everything the Step-A code does survives, amplified:

- **Weighting matrix.** Forty moment contributions, many of them products of heavy-tailed series, make the raw covariance of moment conditions worse-conditioned than Step A's (2.6×10^6 before truncation). Ledoit–Wolf shrinkage remains mandatory; keep the condition-number guardrail (warn above 100, ridge backstop) and expect a larger shrinkage intensity. Consider estimating W once at the empirical moments (as now) and, once stable, a second GMM step at $\hat{\Theta}'$.
- **Analytic Jacobian.** Every moment is a product of (i) a coefficient polynomial in $(\sigma_c^2, S, A, B, \psi, c)$ — linear or bilinear, trivial derivatives — and (ii) a damping factor $f, g, \hat{g}$ evaluated at κ_c or κ_j times an exponential lag term. Chain rule through the composites; the only new scalar derivatives needed are $f', g', \hat{g}'$ (Appendix A) — already in the codebase for f', g' . The mean $\mathbb{E}[\text{TPV}]$ additionally involves $\partial_{(\kappa_j, \psi, c)} cA/(\kappa_j + \psi c)$: elementary quotients.
- **Multi-start.** The surface now has a genuine ridge: $(\psi \uparrow, S \downarrow)$ trade off in C_B while C_F compensates through c . Seed the multi-start grid from the composite mental map above: $\hat{\kappa}_j$ from a log-linear fit to the empirical JV-acov slope, $\hat{C}_B/\hat{S} \rightarrow \psi$, $(\hat{C}_F - \hat{C}_B)/\hat{B} \rightarrow c$ — i.e., *method-of-moments initialization from exactly the diagnostics that motivated the extension*.
- **Uncertainty.** Parameters near bounds and ridge geometry make point estimates insufficient: run the repository's block bootstrap over Θ' and report intervals for the *derived* quantities too (η , b , half-lives $\ln 2/\kappa_j$, $\ln 2/b$) — the branching ratio η with its CI is the headline number of the whole exercise (“how self-exciting is BTC jump risk?”).

Chapter 7

Filtering: projecting the latent states

7.1 LMMSE without Gaussianity

The Step-B objective is the best *linear* projection of $X_t = (V^c(t), V^j(t))'$ onto a window of observables $Y_t = (\text{TPV}_{t-j}, \text{JV}_{t-j})_{j=0}^{p-1}$:

$$\hat{X}_t = \mu_X + \Sigma_{XY} \Sigma_{YY}^{-1} (Y_t - \mu_Y), \quad \text{MSE} = \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX}.$$

No Gaussianity is assumed or needed: among all linear functions of Y_t this minimizes mean-squared error, period. What the model supplies is the *population second-moment structure*; better structure, better filter. This is where the extension pays concretely: under Step A the model's Σ_{XY} said lagged JV was nearly worthless (loadings decaying at $e^{-6.14k}$) — under $\mathcal{M}'$ the loadings decay at $e^{-\kappa_j k}$ with κ_j estimated from data that clearly wants persistence.

7.2 The projection inputs, in closed form

$\mu_X = (\bar{V}^c, \bar{v})'$; μ_Y stacks (5.2). $\Sigma_{XX} = \text{diag}(\sigma_c^2, S)$ (the factors are independent). The cross block, from Lemma 5.1 (third formula) and the kernel of Lemma 5.2 — all with the *uniform* lag law valid for $k \geq 0$:

$$\begin{aligned} \text{Cov}(V^c(t), \text{TPV}_{t-k}) &= \sigma_c^2 \hat{g}(\kappa_c) e^{-\kappa_c k}, & \text{Cov}(V^c(t), \text{JV}_{t-k}) &= 0, \\ \text{Cov}(V^j(t), \text{TPV}_{t-k}) &= S \hat{g}(\kappa_j) e^{-\kappa_j k}, & \text{Cov}(V^j(t), \text{JV}_{t-k}) &= C_F \hat{g}(\kappa_j) e^{-\kappa_j k}. \end{aligned} \quad (7.1)$$

(The V^j –JV entry combines the jump-surprise kernel, $\bar{\lambda} m_{kh^2} = cB$, with the intensity channel, ψS ; their sum is the same C_F as in the forward cross moment — a pleasing and non-accidental coincidence: the filter “sees” jump history through exactly the channel the cross-covariance diagnostic measures.)

Σ_{YY} stacks the full Theorem 5.3 moment set on the chosen window.

7.3 Model, hybrid, and the PSD guardrail

Three ways to fill $(\Sigma_{XY}, \Sigma_{YY})$:

1. **Pure model:** both from $\hat{\Theta}'$. Internally consistent; optimal if $\mathcal{M}'$ is right; still misspecification-fragile.

2. **Hybrid:** Σ_{XY} from the model, Σ_{YY} empirical with Ledoit–Wolf shrinkage. Robust to residual Σ_{YY} misspecification; the pragmatic default. *Guardrail:* the stitched joint matrix $\begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix}$ is not automatically PSD; check its smallest eigenvalue, and if negative, shrink $\Sigma_{XY} \rightarrow \alpha \Sigma_{XY}$ with the largest $\alpha \in (0, 1]$ restoring PSD (report α ; $\alpha \ll 1$ is itself a misspecification alarm). Also verify the implied MSE matrix is PSD — same check, same fix.
3. **Comparison as deliverable:** run both on simulated data from $\mathcal{M}'$ (where the latent truth is known) and report state-recovery RMSE. Under Step A’s $\hat{\Theta}$ the pure-model filter is provably leaving information on the table; *quantifying* that gap is a genuinely interesting result and directly showcases the misspecification-cost theme of the write-up.

Project impact 7.1. Filter outputs $(\hat{V}_t^c, \hat{V}_t^j)$ are the state variables for everything in Chapter 8. Their quality ceiling is set by the covariances (7.1); those are trustworthy only if the moment formulas behind them survived (i) simulation verification (Chapter 9) and (ii) the over-identification tests of Chapter 6. This dependency chain is why estimation rigor comes before any VRP claim.

Chapter 8

The variance risk premium and regime classification

8.1 Physical-measure forecasts in closed form

Total quadratic variation over a horizon τ (in days) is $\text{QV}_{t,t+\tau} = \int_t^{t+\tau} (V^c + V^j) \, ds + \iint h^2 \, d\mu$. Using the conditional means (Theorem 4.2 and its CIR analogue) and Lemma 2.5, with $\tilde{\tau}(a) := (1 - e^{-a\tau})/a$:

$$\mathbb{E}_t^{\mathbb{P}}[\text{QV}_{t,t+\tau}] = \underbrace{(\bar{V}^c + \bar{v} + A)\tau}_{\text{unconditional level}} + (V_t^c - \bar{V}^c) \tilde{\tau}(\kappa_c) + (1 + \psi)(V_t^j - \bar{v}) \tilde{\tau}(\kappa_j). \quad (8.1)$$

Two structural remarks. First, the jump-energy term enters through $\mathbb{E}_t \int \lambda m_{h^2}$, and since λ is affine in V^j it *adds* ψ to the loading on the jump state: elevated jump variance predicts future variance twice — through its own decay *and* through the extra jumps it will cause. Second, the loading decays with $\tilde{\tau}(\kappa_j)$: for a 30-day horizon and a persistent κ_j , jump-state shocks move even month-ahead expected variance materially. Both effects are absent from the Step-A model. In practice, plug the filtered $(\hat{V}_t^c, \hat{V}_t^j)$ into (8.1).

8.2 The premium is affine in the states

Under a standard affine change of measure (intensity scaling $\lambda^{\mathbb{Q}} = \ell \lambda$, mark tilting $G^{\mathbb{Q}}$, and drift adjustments to V^c), the $\mathbb{Q}$ -expectation of QV has the *same functional form* as (8.1) with risk-adjusted coefficients. Hence

$$\text{VRP}_t(\tau) := \mathbb{E}_t^{\mathbb{Q}}[\text{QV}_{t,t+\tau}] - \mathbb{E}_t^{\mathbb{P}}[\text{QV}_{t,t+\tau}] = \alpha_0(\tau) + \alpha_c(\tau) V_t^c + \alpha_j(\tau) V_t^j : \quad (8.2)$$

the variance risk premium is an affine function of the two latent states. This is the theoretical license for the entire classification program: states of the VRP *are* regions of the (V^c, V^j) -plane, so classifying filtered states classifies the premium.

Getting the $\mathbb{Q}$ -side without an option-pricing model. The clean two-step exploits (8.2): obtain a model-free $\mathbb{Q}$ -expected variance from the options market (Deribit DVOL², 30-day), then regress it on the filtered states:

$$\text{DVOL}_t^2 \cdot \frac{30}{365} = \beta_0 + \beta_c \hat{V}_t^c + \beta_j \hat{V}_t^j + \varepsilon_t.$$

The theory says this linear regression is correctly specified (the projection coincides with the structural relation up to filter error), so $\hat{\beta}$ —(the $\mathbb{P}$ -coefficients from (8.1)) estimates the $\alpha(\tau)$ ’s directly. No parametric option model, no risk-price calibration: honest, simple, and defensible in an interview. Expected sign restrictions (β_j exceeding the physical loading $(1 + \psi)\tilde{\tau}(\kappa_j)$, i.e. $\alpha_j > 0$: jump risk carries a premium) become testable hypotheses — report them.

8.3 Regimes and stability, made precise

The affine structure tells you *which* features are sufficient and *what stability means*:

- **Features.** The jump share $\hat{V}_t^j/(\hat{V}_t^c + \hat{V}_t^j)$ and the total level $\hat{V}_t^c + \hat{V}_t^j$ span the state plane; by (8.2) any VRP regime is a half-plane-ish region in these coordinates. Add the *intensity innovation* $JV_t - \mathbb{E}[JV_t \mid \text{history}] = JV_t - (A + \psi(\widehat{V^j\text{-path}} - \bar{v})\text{-term})$: under $\mathcal{M}'$ it is white, so its running size is a real-time misspecification/stress alarm — clustering that even the Hawkes term cannot absorb.
- **Classification.** Start with transparent thresholds on (jump share, level); add a 2–3 state HMM on logs as a comparison; select state count by held-out likelihood, not in-sample BIC alone (545 daily observations). Validate against the known event: 2025-10-10 must be flagged without being told.
- **Stability metrics with model meaning.** (i) Regime persistence compares directly to the structural half-lives $\ln 2/\kappa_c$, $\ln 2/\kappa_j$: a “regime” shorter than the states’ own relaxation times is noise, not regime. (ii) The branching ratio η is a *global* fragility dial: as $\eta \rightarrow 1$ the system approaches criticality — variance of V^j and cluster sizes diverge. Reporting $\hat{\eta}$ with a bootstrap CI, and tracking it over subsamples (2025 vs 2026), turns “stability of the VRP” from a vibe into a parameter. (iii) Posterior entropy of the HMM gives a day-by-day uncertainty measure; spikes flag transitions.

Project impact 8.1. Deliverables enabled by this chapter, in order: the VRP time series (8.2) on real data (needs the DVOL leg); regime-shaded charts of VRP and jump share; the $\hat{\eta}$ criticality dial with CIs; an event study around 2025-10-10 (VRP spike and decay at rate κ_j — a directly testable model prediction: the post-event decay slope in the data *should* match the GMM-estimated κ_j ; verifying it on an event not targeted by any moment is the strongest single validation this project can produce).

Chapter 9

Implementation: simulation, verification, and code changes

9.1 Exact simulation (no discretization)

Two exact schemes; implement the first, keep the second as cross-check.

Ogata thinning. Between jumps, $\lambda(t)$ decays deterministically: from state $V^j(\tau_i)$, $\lambda(t) = \lambda_0 + \lambda_1 V^j(\tau_i) e^{-b(t-\tau_i)}$, which is bounded by its left value. Iterate: (1) set $\bar{\lambda}_{\text{loc}} = \lambda(\tau_i^+)$; (2) propose the next event after an $\text{Exp}(\bar{\lambda}_{\text{loc}})$ wait; (3) accept with probability $\lambda(\text{proposed time})/\bar{\lambda}_{\text{loc}}$ (the decayed, hence smaller, true rate); if rejected, update the bound and repeat; (4) on acceptance draw $x \sim G$, add $k(x)$ to V^j , record $h^2(x)$. This is exact for monotone-between-jumps intensities. Simulate V^c with the standard exact CIR transition (noncentral χ^2) on a fine grid; form daily TPV, JV by integrating/aggregating.

Cluster construction. Draw immigrants as $\text{Poisson}(\lambda_0)$ on $[0, T]$; recursively give each event $\text{Poisson}(\lambda_1 k(x)/b)$ children with ages $\sim \text{Exp}(b)$; stop when a generation is empty (a.s. finite by $\eta < 1$). Statistically identical output, independent code path — ideal for catching bugs in either.

9.2 The verification protocol (non-negotiable)

Before the first GMM run on real data:

1. Fix a test parameter point with meaningful self-excitation (e.g. $\eta \approx 0.6$, κ_j half-life ≈ 4 d, b half-life ≈ 0.3 d).
2. Simulate $\geq 2 \times 10^5$ days; compute all 40 empirical moments.
3. Compare against Theorem 5.3 with batch-means Monte Carlo error bands; every formula must land inside its band. Repeat at $\lambda_1 = 0$ and confirm collapse to the Step-A formulas (Corollary 5.4).
4. Verify the analytic Jacobian against finite differences at random interior points (max relative error $< 10^{-6}$ at sensible step sizes).

5. Only then: run GMM on simulated data and confirm parameter recovery (estimates within bootstrap bands of truth). This end-to-end rehearsal is also Figure 1 of the write-up’s appendix: “the estimator works when the model is true.”

9.3 Concrete changes to the codebase

component	Step A (<code>test.py</code>)	Step A'
data engineering	5-min bars	1-min bars (signature-justified)
jump-robust estimator	tripower, $\min(\cdot, RV)$ cap	same, add MedRV cross-check
theoretical moments	20 moments, 8 params	40 moments, 9 params (Θ')
damping functions	f, g	$f, g, \hat{g}$ and derivatives
weighting	Ledoit–Wolf + ridge	unchanged (more shrinkage)
bounds	κ_c band, ρ_j band	κ_c band, positivity, $b \geq \ln 2/2$
new module	—	simulator + moment verifier
new outputs	3 fit plots	+ JV-acov and two-sided cross fits, tests $C_J S = C_B C_F$, strict gauge
filter (Step B)	—	Σ_{XY} (7.1); hybrid Σ_{YY} ; PSD guard

Table 9.1 is the parameter dictionary; keep it in the repository documentation and in docstrings (the coding standard requires each code block to cite its SDE equivalent — cite equations of this document by number).

Table 9.1: Dictionary: Step-A parameters vs. Step-A' composites.

Step A	Step A'	note
$\mathcal{I}_{h^2}$	$A = \bar{\lambda} m_{h^2}$	$= \mathbb{E}[\text{JV}]$
$\mathcal{I}_{h^4}$	$B = \bar{\lambda} m_{h^4}$	$= \text{Var}(M)$ level
$\mathcal{I}_{kh^2}$	$cB = \bar{\lambda} m_{kh^2}$	cross asymmetry $C_F - C_B$
$\text{Var}(V^j)$	S	free under relaxed gauge
$-\rho_j$	$b = \kappa_j + \psi c$	microscopic decay (transient)
—	$\kappa_j = b(1 - \eta)$	macroscopic decay (persistent)
—	$\psi = \lambda_1 m_{h^2}, \eta = \psi c/b$	self-excitation strength
κ_c, σ_c^2 , level	unchanged	CIR block untouched

Appendix A

Damping functions and derivatives (for the analytic Jacobian)

With $E := e^{-a}$ (evaluate $1 - E$ as `-expm1(-a)` and $a - 1 + E$ as `a + expm1(-a)`):

$$\begin{aligned} f(a) &= \frac{2(a - 1 + E)}{a^2}, & f'(a) &= \frac{2(1 - E)}{a^2} - \frac{4(a - 1 + E)}{a^3}, \\ \hat{g}(a) &= \frac{1 - E}{a}, & \hat{g}'(a) &= \frac{E}{a} - \frac{1 - E}{a^2}, \\ g(a) &= \hat{g}(a)^2, & g'(a) &= 2\hat{g}(a)\hat{g}'(a). \end{aligned}$$

Small- a limits (useful as unit tests): $f(a) \rightarrow 1 - \frac{a}{3}$, $\hat{g}(a) \rightarrow 1 - \frac{a}{2}$, $g(a) \rightarrow 1 - a$. Large- a : $f \sim 2/a$, $\hat{g} \sim 1/a$, $g \sim 1/a^2$. Lagged terms: for $m(a) = g(a)e^{-a(k-1)}$, $m'(a) = [g'(a) - (k-1)g(a)]e^{-a(k-1)}$, and analogously with $\hat{g}$ and lag k ($\hat{m}' = [\hat{g}' - k\hat{g}]e^{-ak}$). Derived-parameter chain rules: with $b = \kappa_j + \psi c$, $\partial b / \partial \kappa_j = 1$, $\partial b / \partial \psi = c$, $\partial b / \partial c = \psi$; with $\bar{v} = cA/b$, $\partial \bar{v} = (A/b, cA \partial(1/b))$ componentwise.

Appendix B

Useful integrals

For $a > 0$, $k \geq 1$ integer:

$$\int_0^1 \int_0^1 e^{-a|u-w|} du dw = f(a), \quad \int_k^{k+1} \int_0^1 e^{-a(s-u)} du ds = g(a)e^{-a(k-1)},$$

$$\int_0^1 \int_0^s e^{-a(s-u)} du ds = \frac{1}{2}f(a), \quad \int_k^{k+1} e^{-as} ds = \hat{g}(a)e^{-ak}, \quad \int_0^\tau e^{-as} ds = \tilde{\tau}(a) = \frac{1 - e^{-a\tau}}{a}.$$

The triangle integral (third) is exactly half the square integral (first): the geometric reason the contemporaneous cross moment carries $f/2$ where the variance carries f .

Appendix C

Notation glossary

$\mu, \tilde{\mu}, \nu$	jump counting measure, compensated measure, compensator
G, m_ϕ	mark distribution (probability); $m_\phi = \int \phi \, dG$
$\lambda_0, \lambda_1, \lambda(t)$	baseline, excitation slope, intensity $\lambda_0 + \lambda_1 V^j(t^-)$
$b = -\rho_j$	microscopic (per-jump) decay rate of V^j
$\eta = \lambda_1 m_k / b$	branching ratio (children per jump); $\eta < 1$
$\kappa_j = b(1 - \eta)$	effective (observable) decay rate of V^j
$\bar{\lambda}, \bar{v}$	stationary mean intensity, stationary mean of V^j
S	$\text{Var}(V^j) = \bar{\lambda} m_k^2 / (2\kappa_j)$
A, B, ψ, c	composites $\bar{\lambda} m_{h^2}, \bar{\lambda} m_{h^4}, \lambda_1 m_{h^2}$, gauge constant
C_F, C_B, C_J	cross-forward, cross-backward, JV-acov coefficients
$f, g, \hat{g}$	damping trio (Lemma 5.1); $g = \hat{g}^2$
M_t, Λ_t	day- t jump surprise, integrated intensity (5.1)
$\mathcal{E}_k$	$e^{-\kappa_j(k-1)}$ (and $\mathcal{E}_k^c$ with κ_c)

Appendix D

Implementation checklist

1. Switch data engineering to 1-minute bars; add MedRV alongside tripower; keep the $TPV \leq$ RV truncation and its warning.
2. Implement $f, g, \hat{g} +$ derivatives with `expm1`; unit-test against the small-/large- a limits of Appendix A.
3. Implement `theoretical_moments_ext( $\Theta'$ )` returning the 40-vector of Theorem 5.3 and its analytic 40×9 Jacobian; unit-test the $\lambda_1 = 0$ collapse against the existing Step-A function.
4. Build the simulator (thinning; cluster construction as cross-check); run the five-step verification protocol of Chapter 9. Do not proceed past a failing band.
5. Extend `empirical_moments` to the 40-moment menu; reuse Ledoit–Wolf weighting with condition-number guardrails.
6. Estimate; report Θ' , derived $(b, \eta, \lambda_0, \bar{v})$ with block-bootstrap CIs; report the consistency tests $C_J S = C_B C_F$ and $S = c^2 B / (2\kappa_j)$; regenerate all three diagnostic figures with the new model-implied curves overlaid.
7. Step B: assemble Σ_{XY} from (7.1), hybrid Σ_{YY} , PSD guardrail; simulation study comparing pure-model vs hybrid filters under both $\hat{\Theta}$ (old) and $\hat{\Theta}'$ (new).
8. VRP: source DVOL; regress on filtered states per Chapter 8; build the VRP series, regimes, $\hat{\eta}$ dial, and the 2025-10-10 event study (test: post-event decay slope = $\hat{\kappa}_j$).